

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED

BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-12	Team No-17	Project Title-Inventory & Warehouse Management System
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1. Abstract

The Inventory and Warehouse Management System is a web-based application designed to efficiently manage products, warehouses, suppliers, staff, stock transactions, and orders through a centralized platform. The system provides a secure and organized environment for monitoring inventory, tracking stock movement, managing warehouse capacity, and viewing products stored in each warehouse. It uses HTML, CSS, and JavaScript for the frontend, Python FastAPI for the backend, and MySQL for database management, with authentication and role-based access control to protect the system.

The system also focuses on improving warehouse efficiency through Warehouse Slotting Optimization. DSA techniques such as sorting, priority queues, and greedy algorithms can be used to prioritize products based on demand and assign them to suitable warehouse locations considering factors such as storage capacity, product requirements, and accessibility. This helps improve inventory visibility, utilize warehouse space efficiently, and reduce the time and distance required for product retrieval.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090196	M. Laasya Sri	M. Laasya Sri
2520090197	M. Rajini	M. Rajini
2520090313	K. Ramya	K. Ramya

3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

Frontend: React.js, Chart.js

Backend: Node.js, Express.js

Database: MySQL

APIs / Tools: Barcode Scanner API, JWT (Authentication)

Guide Name and Signature (remarks if any) _____